

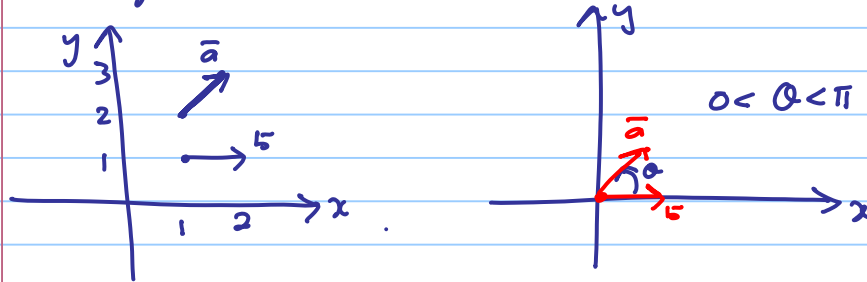
- ① In unit 6.2, we defined the product of a scalar with $\vec{a}$.
 $c\vec{a} = c\langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$.
- ② In this unit we will define the dot product between 2 vectors: $\vec{a} \cdot \vec{b}$
- ③ In unit 6.4 we will define the cross product: $\vec{a} \times \vec{b}$.

Definition: p. 807.

Properties of the dot product: p. 807

Geometric interpretation in terms of the angle θ between vectors $\vec{a}$ and $\vec{b}$ (see p. 808).

The angle between 2 vectors $\vec{a}$ and $\vec{b}$ is the angle between the representations that start at the origin and is always between 0 and π



Theorem 3 (p. 808)

If θ is the angle between $\vec{a}$ and $\vec{b}$, then

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

and: if $\vec{a} \neq \vec{0}$, $\vec{b} \neq \vec{0}$, then $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$.
 (p. 809)

Orthogonal vectors: angle between them is $\pi/2$. (p. 809)

(NB p. 809) Theorem: $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$.

Proof:

$$\textcircled{1} (\Rightarrow) \text{ If } \vec{a} \perp \vec{b}, \text{ then } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \frac{\pi}{2} = 0$$

$$\textcircled{2} (\Leftarrow) \text{ If } \vec{a} \cdot \vec{b} = 0, \text{ then } \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = 0 \Rightarrow \theta = \frac{\pi}{2}$$

Some examples:

$$\begin{aligned} \textcircled{1} \quad \langle -2, 3 \rangle \cdot \langle 1, 2 \rangle &= (-2)(1) + (3)(2) \\ &= -2 + 6 \\ &= 4. \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad (8\vec{i} + 2\vec{j} - \vec{k}) \cdot (2\vec{i} + 3\vec{j}) &= \langle 8, 2, -1 \rangle \cdot \langle 2, 3, 0 \rangle \\ &= 16 + 6 + 0 \\ &= 22. \end{aligned}$$

$$\textcircled{3} \quad \langle 2, 3, 4 \rangle \cdot \vec{k} = \langle 2, 3, 4 \rangle \cdot \langle 0, 0, 1 \rangle = 0 + 0 + 4 = 4.$$

$\textcircled{4}$ Find k such that $\vec{a} = \langle -1, 3, k \rangle$ and $\vec{b} = \langle 2, k, 4 \rangle$ are orthogonal.

We need: $\vec{a} \cdot \vec{b} = 0$

$$\Rightarrow \langle -1, 3, k \rangle \cdot \langle 2, k, 4 \rangle = 0$$

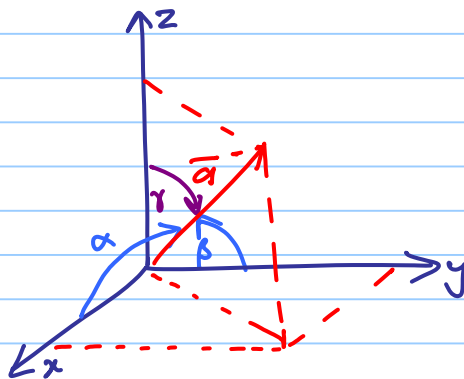
$$\Rightarrow -2 + 3k + 4k = 0$$

$$\Rightarrow 7k = 2$$

$$\Rightarrow \underline{k = 2/7.}$$

Direction angles and direction cosines: (p. 810)

The direction angles of $\vec{a}$ are the angles α, β, γ that $\vec{a}$ makes with the x -, y - and z -axis, respectively.



Direction angles: α, β, γ

Direction cosines (p. 810)

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and the direction angles are α, β, γ , then:

$$\vec{a} = \langle |\vec{a}| \cos \alpha, |\vec{a}| \cos \beta, |\vec{a}| \cos \gamma \rangle$$

$$\Rightarrow a_1 = |\vec{a}| \cos \alpha \Rightarrow \cos \alpha = \frac{a_1}{|\vec{a}|}$$

$$\text{and } a_2 = |\vec{a}| \cos \beta \Rightarrow \cos \beta = \frac{a_2}{|\vec{a}|}$$

$$\text{and } a_3 = |\vec{a}| \cos \gamma \Rightarrow \cos \gamma = \frac{a_3}{|\vec{a}|}$$

Example: see next page.

Example: Find the direction angles and the direction cosines of $\vec{v} = 2\vec{i} - 4\vec{j} + 4\vec{k}$.

$$\vec{v} = \langle 2, -4, 4 \rangle.$$

$$|\vec{v}| = \sqrt{4 + 16 + 16} = \sqrt{36} = 6.$$

$$\cos \alpha = \frac{2}{6} = \frac{1}{3} \quad \text{and} \quad \alpha = \arccos \frac{1}{3}$$

$$\cos \beta = \frac{-4}{6} = -\frac{2}{3} \quad \beta = \arccos -\frac{2}{3}$$

$$\cos \gamma = \frac{4}{6} = \frac{2}{3} \quad \gamma = \arccos \frac{2}{3}.$$

$$\alpha, \beta, \gamma \in [0, \pi].$$